

1) In the 2021 baseball season, the Brewers runs per game, B , varied from game to game but had $\mu_B = 3.7$ and $\sigma_B = 1.1$. The Cubs runs per game, C , were $\mu_C = 4.9$ and $\sigma_C = 1.6$.

Find the following (show your work):

μ_{B+C}	μ_{B-C}
σ_{B+C}	σ_{B-C}

2) At our town's ice cream shop, if you order a cone, you can determine the number of scoops you would like. Let X represent the number of scoops ordered and the probability that a random customer would ask for that number of scoops.

X	1	2	3	4	5
$P(X)$	0.20	0.38	0.28	0.11	0.03

It costs \$2.50 for the cone and \$0.75 for each scoop of ice cream. Find the amount of money the store expects to make on a randomly selected customer.

3) Suppose the owners of the ice cream shop want to keep track of their profits. The cost of a cone is \$0.10 for the shop and the cost of a scoop of ice cream is \$0.45. What is the expected profit on a randomly selected customer? Use the information from problem #2 to help you answer this question. Show your work.